

RF-100 ExecutionProof Governance Standard

Verification-First Execution Control for High-Consequence Systems

Version 0.9 · DRAFT FOR REVIEW · July 2026

Built on RF-000 (Proof Before Power™ Doctrine, Published)

Includes Annex A: RF-100:AI Conformance Profile (v0.9 Draft)

**DRAFT — v0.9 — NOT YET RELEASED FOR CONFORMANCE CLAIMS —
SUBJECT TO CHANGE**

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1. Scope

This standard defines the minimum normative structure that a high-consequence system **MUST** satisfy before executing a governed action. It applies to any system — AI agent, automated workflow, API, human-authorized process, financial system, or infrastructure controller — whose actions can produce material, irreversible, or safety-relevant consequences.

RF-100 governs the **execution boundary**: the point at which a proposed action transitions from intention to effect. It does not govern model training, application design, or post-execution remediation, except insofar as those produce evidence consumed at the boundary.

2. Normative Language

The key words **MUST**, **MUST NOT**, **SHOULD**, **SHOULD NOT**, and **MAY** in this document are to be interpreted as described in RFC 2119 / RFC 8174. Requirements are individually numbered (RF-100.x.y) for traceability in conformance assessment.

3. Definitions

Term	Definition
Governed Action	An action designated by the adopting organization as subject to this standard, by inclusion in an Execution Boundary Profile.
Execution Boundary	The control point at which a proposed Governed Action is verified before it is permitted to execute.
Actor	The entity (human, service, AI agent, or workflow) proposing a Governed Action.
Verification Gate	The mechanism that evaluates a proposed action against the ten RF-100 elements and returns a decision.
Decision	One of ALLOW, HOLD, or DENY, rendered by the Verification Gate prior to execution.
ProofRecord™	A signed, tamper-evident record of a verification decision and its inputs, generated before execution proceeds.
Execution Boundary Profile (EBP)	The documented configuration for a Governed Action: authority rules, evidence requirements, policy constraints, state conditions, and risk thresholds.
Evidence Freshness	The maximum permissible age of an evidence artifact at decision time.

4. Conformance Model

A system claims conformance *per Governed Action*, not globally. A conforming implementation:

- **MUST** satisfy all MUST-level requirements in Sections 5.1–5.10 for each Governed Action;
- **SHOULD** satisfy all SHOULD-level requirements, and **MUST** document justification for any deviation;
- **MUST** maintain a current Execution Boundary Profile for each Governed Action;

- **MUST NOT** claim RF-100 conformance for actions that bypass the Verification Gate under any operating condition, including degraded mode.

Draft-stage note: at v0.9, no certification pathway exists (RF-102 is planned). Organizations MAY claim "designed in alignment with RF-100 v0.9 draft" but MUST NOT claim certification.

5. The Ten Elements — Normative Requirements

5.1 Actor Identity

RF-100.1.1 — Every proposed Governed Action **MUST** be attributable to exactly one identified Actor via a cryptographically verifiable credential valid at decision time.

RF-100.1.2 — Anonymous, shared, or unattributable credentials **MUST NOT** be accepted for Governed Actions.

RF-100.1.3 — Where an Actor acts on behalf of another principal (delegation), the full delegation chain **MUST** be resolvable and recorded.

RF-100.1.4 — AI and automated Actors **SHOULD** carry identity distinct from the human or service that deployed them.

5.2 Authority

RF-100.2.1 — The Verification Gate **MUST** confirm that the Actor holds current, explicit authority for the specific Governed Action, evaluated at decision time. Cached or previously granted authority **MUST NOT** be relied upon beyond its documented validity window.

RF-100.2.2 — Capability **MUST NOT** be treated as authority. Possession of credentials or technical access alone is insufficient.

RF-100.2.3 — An Actor **MUST NOT** approve its own proposed action (self-approval blocking). Where multi-party approval (M-of-N) is configured, approvers **MUST** be distinct Actors from the proposer.

RF-100.2.4 — Authority revocation **SHOULD** take effect at the Verification Gate within a documented propagation time, which **MUST** be stated in the EBP.

5.3 Policy

RF-100.3.1 — Each Governed Action **MUST** be evaluated against a version-controlled, machine-evaluable policy set defined in the EBP.

RF-100.3.2 — The policy version applied to each decision **MUST** be recorded in the ProofRecord™.

RF-100.3.3 — Policy changes **MUST** follow a documented change-control process with distinct author and approver, and **SHOULD** be testable in simulation before enforcement.

5.4 Evidence

RF-100.4.1 — The EBP **MUST** enumerate the evidence artifacts required for the Governed Action, including source, format, and freshness limit for each.

RF-100.4.2 — Evidence exceeding its freshness limit at decision time **MUST** be treated as absent.

RF-100.4.3 — Missing or stale required evidence **MUST** result in HOLD or DENY, never ALLOW.

RF-100.4.4 — Evidence artifacts **SHOULD** be integrity-protected (hashed or signed) from source to gate.

5.5 System State

RF-100.5.1 — The EBP **MUST** define the system-state conditions under which the Governed Action is admissible (e.g., operational mode, maintenance windows, dependency health, capacity limits).

RF-100.5.2 — State **MUST** be sampled at decision time; assumed or default state values **MUST NOT** substitute for sampled state.

RF-100.5.3 — If state cannot be determined, the Gate **MUST** return HOLD or DENY (fail-closed).

5.6 Risk

RF-100.6.1 — The EBP **MUST** define quantitative or categorical risk thresholds for the Governed Action (e.g., value limits, rate limits, blast-radius bounds, anomaly scores).

RF-100.6.2 — Actions exceeding a risk threshold **MUST NOT** receive ALLOW; they **MUST** receive HOLD (for escalation) or DENY.

RF-100.6.3 — Cumulative and velocity risk (aggregation across recent actions) **SHOULD** be evaluated, not only per-action risk.

5.7 Decision

RF-100.7.1 — The Verification Gate **MUST** return exactly one of: **ALLOW**, **HOLD**, or **DENY**, prior to any execution effect.

RF-100.7.2 — The decision sequence is invariant: verification **MUST** complete before execution begins. The sequence **MUST NOT** be reversed, bypassed, or deferred, including in degraded, emergency, or high-load conditions. Emergency procedures **MUST** themselves be Governed Actions with their own EBPs.

RF-100.7.3 — **HOLD** **MUST** have a defined resolution path (approver set, timeout, and timeout disposition). A **HOLD** that times out **MUST** resolve to **DENY** unless the EBP explicitly documents otherwise.

RF-100.7.4 — Gate unavailability **MUST** fail closed for Governed Actions: no gate, no execution.

RF-100.7.5 — Decision latency **SHOULD** be measured and reported per Governed Action.

5.8 Proof Record

RF-100.8.1 — Every decision (**ALLOW**, **HOLD**, and **DENY**) **MUST** generate a tamper-evident, cryptographically signed record before the action proceeds, containing at minimum: actor identity and delegation chain, authority basis, policy version, evidence references and freshness status, sampled state, risk evaluation, decision, timestamp, and gate identity.

RF-100.8.2 — Proof records **MUST** be retained per a documented retention policy and protected from modification and deletion within the retention window.

RF-100.8.3 — Proof records **SHOULD** be independently verifiable without access to the live gate (offline verification of signatures and chain integrity).

5.9 Execution Outcome

RF-100.9.1 — The actual execution outcome (success, failure, partial completion) **MUST** be captured and linked to the originating proof record.

RF-100.9.2 — Divergence between the verified action and the executed effect (scope creep, parameter mutation post-decision) **MUST** be detectable and **SHOULD** raise an alert.

RF-100.9.3 — An **ALLOW** decision **MUST NOT** be reusable for more than one execution instance unless the EBP explicitly defines a bounded reuse window and count.

5.10 Review Path

RF-100.10.1 — Each Governed Action **MUST** have a documented review path: who reviews decisions and outcomes, at what cadence, and with what escalation authority.

RF-100.10.2 — **DENY** and timed-out **HOLD** decisions **SHOULD** be reviewed within a documented interval to detect misconfiguration, attack patterns, or authority gaps.

RF-100.10.3 — The complete decision trail for any Governed Action **MUST** be reconstructable from proof records alone.

6. Execution Boundary Profile — Required Contents

Each Governed Action's EBP **MUST** document: (a) action definition and scope; (b) permitted Actor classes and identity requirements; (c) authority rules including approval structure and revocation propagation time; (d) policy set and version; (e) required evidence with freshness limits; (f) admissible state conditions; (g) risk thresholds; (h) HOLD resolution path and timeout disposition; (i) proof record retention period; (j) review cadence and owner. The EBP itself **MUST** be version-controlled.

Annex A — RF-100:AI Conformance Profile (v0.9 Draft)

This profile specializes RF-100 for **AI agents and autonomous systems**: systems in which a machine-learning-driven Actor proposes actions with limited or no human review per action.

A.1 Additional definitions

Term	Definition
Agent	An AI system capable of proposing or initiating Governed Actions via tools, APIs, or effectors.
Tool Call	An agent-initiated invocation of an external capability. Tool calls with material consequence are Governed Actions.
Intent Record	The agent's stated objective and reasoning context accompanying a proposed action.

A.2 Profile requirements

RF-100:AI.1 — Every agent **MUST** hold its own Actor identity, distinct from its deploying human, service account, and any other agent instance. Agent identity **MUST** include model/version and deployment context.

RF-100:AI.2 — Agent authority **MUST** be scoped per tool and per action class, not granted globally. "The agent has an API key" is capability, not authority (per RF-100.2.2).

RF-100:AI.3 — An agent **MUST NOT** approve, resolve HOLDS for, or modify the EBP of any action it proposed — including via a second agent instance under common control (extended self-approval blocking).

RF-100:AI.4 — Proposed agent actions **MUST** carry an Intent Record, which **MUST** be included in the ProofRecord™. The gate **SHOULD** evaluate consistency between stated intent and the action's actual parameters.

RF-100:AI.5 — Agent-proposed actions above a documented consequence threshold **MUST** route to HOLD for human or M-of-N resolution. The threshold **MUST** be defined in the EBP and **SHOULD** be reviewed as agent behavior data accumulates.

RF-100:AI.6 — Velocity limits **MUST** be enforced per agent identity (actions per interval, cumulative value per interval), and breaches **MUST** fail to HOLD or DENY.

RF-100:AI.7 — Prompt- or input-injection indicators **SHOULD** be treated as risk signals at the gate; actions proposed during flagged sessions **SHOULD NOT** receive ALLOW without elevated review.

RF-100:AI.8 — Chained agent workflows (agent A invokes agent B) **MUST** preserve the full delegation chain in each proof record; each consequential hop is independently a Governed Action.

RF-100:AI.9 — Model or prompt updates to a deployed agent **SHOULD** trigger re-evaluation of the agent's EBPs and **MUST** update the agent's recorded identity version.

A.3 Minimal conformance for a first pilot boundary

A single-boundary pilot claims draft alignment when, for one Governed Action: agent identity (AI.1), scoped authority (AI.2), self-approval blocking (AI.3), intent capture (AI.4), a consequence threshold with HOLD routing (AI.5), and signed proof records (RF-100.8.1) are all operational, with fail-closed behavior (RF-100.7.4) demonstrated by test.

Annex B — Revision History & Status

Version	Date	Status
v0.9	July 2026	Draft for review and pilot alignment. Normative numbering stable; requirements subject to change based on pilot feedback. RF-101 (assessment) and RF-102 (certification) not yet available; no certification claims permitted.

Disclosures. This draft standard is published by Remnant Fieldworks Inc. as part of a demo-stage standards pathway under active development. It should be evaluated through direct pilot, technical, legal, and compliance review. U.S. patent applications pending; no assurance is given that any claim will be granted. This document does not constitute legal advice. Remnant Fieldworks™, Proof Before Power™, Verification Before Execution™, ExecutionProof™, and ProofRecord™ are trademarks or pending trademarks of Remnant Fieldworks Inc. © 2026 Remnant Fieldworks Inc. All rights reserved.